

① model: $L_{\text{eff}}^{\lambda\lambda'} = \sqrt{f_{\text{eff}}^{\lambda\lambda'}} d_{\alpha}^{\dagger} d_{\beta}$ $\mu, \lambda, \lambda' = \{\alpha, \beta\}$ $d = \{\alpha, \beta\}$

$\Gamma_{k\ell}^{\lambda\lambda'}$ $\Gamma_{k\ell}^{\mu\nu} = \mathcal{K} e^{-\left(\frac{E_{\mu\nu} - E_{\ell\ell}}{\omega_c}\right)^2} \frac{1}{1 + e^{\frac{E_{\mu\nu} - E_{\ell\ell}}{T}}} \rightarrow \text{DE}$

$\mu(k) \leftarrow \nu(p)$ $\uparrow$ shallow.

$$\frac{\Gamma_{k\ell}^{\mu\nu}}{\Gamma_{p\ell}^{\nu\mu}} = \frac{\cancel{\mathcal{K}} e^{-\left(\frac{E_{\mu\nu} - E_{\ell\ell}}{\omega_c}\right)^2}}{\cancel{\mathcal{K}} e^{-\left(\frac{E_{\nu\mu} - E_{\ell\ell}}{\omega_c}\right)^2}} \cdot \frac{1 + e^{\frac{E_{\nu\mu} - E_{\ell\ell}}{T}}}{1 + e^{\frac{E_{\mu\nu} - E_{\ell\ell}}{T}}} = \frac{1 + e^x}{1 + e^{-x}} = e^{-x} = e^{-(E_{\mu\nu} - E_{\ell\ell})/T}$$

$\rightarrow \Gamma_{k\ell}^{\mu\nu} = \Gamma_{p\ell}^{\nu\mu} \cdot e^{-(E_{\mu\nu} - E_{\ell\ell})/T}$ $\left. \begin{array}{l} \mu(k) \leftarrow \nu(p) \\ \mu(k) \rightarrow \nu(p) \end{array} \right\}$

② model

$\Gamma_{k\ell}^{\mu\nu} = \mathcal{K}^{\mu\nu} e^{-\left(\frac{\Delta E_{\ell}}{\omega_c}\right)^2} f\left(\frac{E_{\mu\nu} - \mu}{T}\right) f\left(-\frac{E_{\ell\ell} - \mu}{T}\right)$

$1 - f(E) = f(-E)$

$\frac{1}{1 + e^x} = \frac{e^{-x}}{1 + e^{-x}} = 1 - \frac{1}{1 + e^{-x}}$

DB

$\mathcal{K}^{\mu\nu} = \mathcal{K}^{\nu\mu}$

$$\frac{\Gamma_{k\ell}^{\mu\nu}}{\Gamma_{p\ell}^{\nu\mu}} = \frac{f(x) f(-y)}{f(-x) f(y)} = \frac{\frac{1}{1 + e^x} \cdot \frac{1}{1 + e^{-y}} \cdot \frac{e^y}{e^x}}{\frac{1}{1 + e^{-x}} \cdot \frac{1}{1 + e^y} \cdot \frac{e^x}{e^y}} = e^{-(x-y)} = e^{-(E_{\mu\nu} - \mu - E_{\ell\ell} + \mu)/T} = e^{-\Delta E_{\ell}/T} \checkmark$$

$x = \frac{E_{\mu\nu} - \mu}{T}$ $y = \frac{E_{\ell\ell} - \mu}{T}$

Thermalization at Low T via Weakly damped Multi site Baths:

goal: $\{c_i\}$, in polylog hopel $\sim G_E$, for f.k.m. T_B in pos:

$\rightarrow \beta_B = \frac{1}{T} e^{-(H_B - \mu_B)/T_B}$ $\# \text{ bath degrees} \rightarrow \log \beta_B = 0$

impose DB in spectral H.B.

noninteracting/diagonalizable system:

$H_B \rightarrow J^B = V^{\dagger} \varepsilon^B V$ $\varepsilon^B = \text{diag}(\varepsilon_1^B, \dots, \varepsilon_n^B)$ $H_0 = \sum_k c_k^{\dagger} c_k = \sum_k V^{\dagger} c_k^{\dagger} \varepsilon_k^B V$

$\{c_k, c_k^{\dagger}\} = \delta_{kk}$

$\rightarrow \beta_B = \prod_{k=1}^n \left(f_k c_k^{\dagger} c_k + (1 - f_k) c_k c_k^{\dagger} \right)$ $f_k = \frac{1}{1 + e^{(\varepsilon_k - \mu)T}}$

$$\rightarrow \underline{L_k^{\text{in}} = \sqrt{\gamma} f_k a_k^+ \quad L_k^{\text{out}} = \sqrt{\gamma(1-g_k)} c_k} \quad \left. \begin{array}{l} A^\dagger + A \\ B^\dagger + B \end{array} \right\} \text{ taki operatori su geturip.}$$

$$\underline{Z_B f_B = 0!}$$

(3a) bosonski bath:

$$H_B = \sum_k \omega_k b_k^\dagger b_k, \quad \beta \approx e^{-\beta H_B}, \quad \bar{n}_B = \frac{1}{e^{\omega_k/T} - 1}$$

$$L_z^- = \sqrt{\gamma_k(\bar{n}_k + 1)} b_k, \quad L_z^+ = \sqrt{\gamma_k \bar{n}_k} b_k^\dagger \rightarrow L_{zB}^{++}, L_{zB}^{+-}, L_{zB}^{-+}, L_{zB}^{--} = L_z^{(\pm)} L_B^{(\pm)}$$

$$\mathcal{D}[\cdot] = \left\{ L_j \rho L_j^\dagger - \frac{1}{2} \{ L_j^\dagger L_j, \rho \} \right\}$$

① QN $\hookrightarrow$ od matrice:

$$H_S = \sum_k (a_k^\dagger, b_k^\dagger) \begin{pmatrix} \epsilon_a(k) + V n_k & -\Delta \\ -\Delta & \epsilon_b(k) + V n_k \end{pmatrix} \begin{pmatrix} a_k \\ b_k \end{pmatrix}^{\text{tr}}$$

$$\rightarrow \Delta = \sum_k \frac{V}{N_k} \langle a_k^\dagger b_k \rangle \rightarrow U_k = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \quad \psi_k = U_k d_k; \quad d_k = \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix}$$

$$\rightarrow H_S = \sum_k d_k^\dagger U_k^\dagger h_k U_k d_k = \sum_{\substack{k \\ \lambda \in \{A, B\}}} E_{\lambda k} d_{\lambda k}^\dagger d_{\lambda k}$$

formi:

$$\rightarrow H_B = \sum_z \omega_z f_z^\dagger f_z$$

$$\rightarrow \text{d.ph.} \quad H_{SB} = \lambda \sum_{k,z} \psi_{k+z}^\dagger \begin{pmatrix} g_a(k,z) & 0 \\ 0 & g_b(k,z) \end{pmatrix} \psi_k (f_z + f_{-z}^\dagger) =$$

$$= \lambda \sum_{\substack{k,z \\ \lambda, \lambda'}} d_{\lambda, k+z}^\dagger \underbrace{(U_{k+z}^\dagger g_{k+z} U_k)}_{[G_{k,z}]_{\lambda, \lambda'}} d_{\lambda', k} (f_z + f_{-z}^\dagger) =$$

$[G_{k,z}]_{\lambda, \lambda'} \rightarrow \{ \alpha \rightarrow \alpha, \alpha \rightarrow \beta \text{ in } \mathbb{P} \rightarrow \mathbb{P} \text{ mapping} \}$

$$= \lambda \sum_{\substack{k,z \\ \lambda, \lambda'}} d_{\lambda, k+z}^\dagger [G_{k,z}]_{\lambda, \lambda'} d_{\lambda', k} (f_z + f_{-z}^\dagger) \quad \otimes$$

$$f_B = \frac{e^{-H_B/T}}{Z} \quad , \quad N_z = \frac{1}{e^{\omega_z/T} - 1} \quad , \quad \langle f_z^\dagger f_z \rangle_{f_B} = N_z \quad , \quad \langle f_z f_z^\dagger \rangle_{f_B} = N_z + 1$$

$$Z_B f_B \propto \sum_z (N_z + 1) \mathcal{D}[b_z] \rho + N_z \mathcal{D}[b_z^\dagger] \rho \stackrel{?}{=} 0$$

Transition: $\lambda(l) \rightarrow \lambda(l+\nu)$, $E_{ij} = E_{l, l+\nu} - E_{l, l}$ at λ_5 ?

Fermi:

$$H \sim \sum_{i,j} d_i^\dagger d_j G_{ij} (f_z + f_z^\dagger)$$

$$d_i^\dagger d_j(t) = e^{iH_0 t} d_i^\dagger d_j e^{-iH_0 t} = e^{iE_{ij} t} d_i^\dagger d_j$$

$$f_z(t) = e^{-i\omega_z t} f_z$$

$$f_z^\dagger(t) = e^{i\omega_z t} f_z^\dagger \quad (\omega_z = \omega_{-z})$$

$$L_{ij}^{abs} = L_{ij}^a = \sqrt{2\pi\lambda^2 |G_{ij}|^2 N_z \delta(\Delta E_{ij} - \omega_z)} d_i^\dagger d_j = \sqrt{\gamma_{ij}^a} d_i^\dagger d_j$$

$$L_{ij}^{em} = L_{ij}^e = \sqrt{2\pi\lambda^2 |G_{ij}|^2 (N_z + 1) \delta(\Delta E_{ij} + \omega_z)} d_i^\dagger d_j = \sqrt{\gamma_{ij}^e} d_i^\dagger d_j$$

$$\gamma = \gamma(N_z(t)) \rightarrow \gamma(1)$$

$$\mathcal{Q}[l] = \sum_{i \neq j} \mathcal{Q}(l_{ij}) p$$

$$l = (\lambda, k)$$

GGE toh: $n_e = \sum_{ij} \left\langle L_{ij}^{t\dagger} n_e L_{ij}^t - \frac{1}{2} \{L_{ij}^{t\dagger} L_{ij}^t, n_e\} \right\rangle_{GGE} =$ (analog to p. 11)

$$\Gamma = \gamma(\tau_1) + \gamma(\tau_2)$$

$$= (1 - n_e) \sum_{j \neq i} \Gamma_{e=j}^t n_j - n_e \sum_{j \neq i} \Gamma_{j=e}^t (1 - n_j)$$

$$GGE: f(t) = \frac{1}{Z} e^{-\sum_k \mu_k n_k}$$

$$\Leftrightarrow n_k = \frac{1}{e^{\mu_k} + 1} \Leftrightarrow \frac{\partial n_k}{\partial \mu_k} = \frac{-e^{\mu_k}}{(1 + e^{\mu_k})^2} = -n_k^2 e^{\mu_k} =$$

$$\rightarrow \mu_k = - \frac{\dot{n}_k}{n_k(1 - n_k)} = - \dot{n}_k \frac{(1 + e^{\mu_k})^2}{e^{\mu_k}}$$

$$e^{\mu_k} \Theta^2 (e_2 \text{ to } v)$$

clausen

3.4.1, 1m, script

$$= -n_k \cdot \left(\frac{1}{e^{\mu_k} + 1} e^{\mu_k} \right) =$$

$$= -n_k \left(\frac{1}{e^{\mu_k} + 1} \right) =$$

$$= -n_k \left(1 - \frac{1}{e^{\mu_k} + 1} \right) =$$

$$= -n_k (1 - n_k)$$